

OR504 Final Project: Portfolio Optimization with CAPM

Max Crisalli, Paras Gupta, Partha Sanampudi, Ryan Landey

April 28, 2026

Abstract

Portfolio optimization remains a focal point in financial decision-making, where investors seek to balance expected return and risk when allocating capital across assets. In this study, we develop a multi-objective portfolio optimization model based on the Capital Asset Pricing Model (CAPM), where risk is measured using beta. The problem is formulated as a mixed-integer goal programming model, incorporating practical constraints such as budget limitations, diversification requirements, and cardinality restrictions. This proposed model is implemented in SAS using PROC OPTMODEL and applied to a dataset of selected assets, including stocks and bond exchange-traded funds (ETFs). The results provide insight into the trade-offs between return and systematic risk and demonstrate how different parameter choices affect the resulting portfolio. Implications beyond this study are discussed as well.

1 Introduction

Investors are faced with the challenge of allocating capital across a set of assets in a way that balances expected return and risk. Developing quantitative methodology to address this trade-off is a central problem in financial mathematics and operations research. Traditional approaches, such as the Markowitz approach, measure risk using covariances. However, systematic risk, as measured by beta in the Capital Asset Pricing Model (CAPM), provides a more interpretable and market-relevant measure of risk. Furthermore, for this project in particular, it makes it easier to solve, since the problem becomes linear as opposed to non-linear.

The report will be organized as follows: problem description and mathematical formulation, implementation and computational results, and key findings and potential applications beyond this model.

2 Problem Description and Mathematical Formulation

2.1 Problem Description

We consider a set of 25 risky assets, along with a risk-free asset, from which the portfolio is constructed. The risky assets include 20 individual stocks and 5 bond exchange-traded funds (ETFs), chosen to provide exposure across multiple sectors and levels of risk. The final portfolio is selected from this set subject to a cardinality constraint that limits the number of chosen assets to at most 15.

The equity portion of the asset set includes large-cap stocks from a variety of industries. These include technology companies such as Apple, Microsoft, NVIDIA, Google, and Amazon; financial institutions including JPMorgan Chase, PNC Financial, Goldman Sachs, Morgan Stanley, and BlackRock; healthcare companies such as UnitedHealth Group, Johnson & Johnson, Pfizer, and Merck; energy firms including ExxonMobil and Chevron; and additional industrial and consumer firms such as Caterpillar, Coca-Cola, Procter & Gamble, and Boeing. This selection provides diversification across major sectors of the economy while maintaining exposure to high-return assets.

The fixed-income portion consists of five bond ETFs: UTEN, AGG, HYG, TIP, and EMB. These funds provide exposure to different segments of the bond market, including U.S. Treasury securities, investment-grade bonds, high-yield bonds, inflation-protected securities, and emerging market debt. Including these assets helps reduce overall portfolio risk and adequately diversify the portfolio.

In addition to the risky assets, a risk-free asset is included, represented by a 3-month U.S. Treasury bill. This asset provides a baseline return with negligible risk and allows the model to allocate capital to a safe investment when appropriate.

Together, this set of assets allows the model to capture a realistic trade-off between return and systematic risk while maintaining diversification across asset classes and sectors.

Furthermore, the objectives of maximizing return and minimizing risk are conflicting, motivating the use of multi-objective optimization methods. So, in this section, we formulate a portfolio optimization problem using goal programming framework with specified target beta and return levels. Deviation variables are introduced to measure the extent to which these targets are not met, and a weighted objective function is used to prioritize the competing goals. The model also incorporates practical constraints, including budget constraints, cardinality limits, diversification bounds, and minimum investment thresholds.

2.2 Sets

$i \in \text{Assets}$	set of all risky assets ($i = 1, \dots, 25$)
$i \in \text{Bonds}$	set of bond ETFs ($i = 21, \dots, 25$)

2.3 Parameters

r_i	is the expected return of asset i
β_i	is the beta value (systematic risk) of asset i
r_f	is the risk-free rate of return
B	is the total available investment budget
R_0	is the target portfolio return
β_0	is the target portfolio beta
K	is the maximum number of assets allowed in the portfolio
L	is the minimum investment in an asset if selected
M	is a sufficiently large constant
w_R	is the weight assigned to return deviation
w_β	is the weight assigned to beta deviation

2.4 Parameter Values

The following parameter values are used in the computational implementation:

$B = 100,000$	(total investment budget)
$K = 15$	(maximum number of selected assets)
$L = 1,000$	(minimum investment per selected asset)
$M = 100,000$	(large constant for linking constraints)

2.5 Decision Variables

$x_i \geq 0$	is the amount invested in asset i .
$x_f \geq 0$	is the amount invested in the risk-free asset
$y_i \in \{0, 1\}$	is 1 if asset i is selected, and 0 otherwise

2.6 Deviation Variables

$d_R^-, d_R^+ \geq 0$	are the under- and over-achievement of target return.
$d_\beta^-, d_\beta^+ \geq 0$	are the under- and over-achievement of target beta.

2.7 Objective Function and Constraints

Minimize Z , or the weighted undesirable deviations from the target return and target beta, respectively:

$$\min Z = w_R d_R^- + w_\beta d_\beta^+ \quad (1)$$

Return Goal Constraint

$$\sum_{i=1}^{25} r_i x_i + r_f x_f + d_R^- - d_R^+ = R_0 \quad (2)$$

The portfolio return should meet the target return.

Beta Goal Constraint

$$\sum_{i=1}^{25} \beta_i x_i + d_\beta^- - d_\beta^+ = \beta_0 \quad (3)$$

The portfolio beta should not exceed the target beta.

Budget Constraint

$$\sum_{i=1}^{25} x_i + x_f = B \quad (4)$$

Total investment must equal the available budget.

Diversification Constraint

$$x_i \leq 0.10B \quad \forall i \in \text{Assets} \quad (5)$$

No more than 10% of the budget is invested in any single asset.

Bond Allocation Constraint

$$\sum_{i \in \text{Bonds}} x_i \geq 0.20B \quad (6)$$

At least 20% of the portfolio must be invested in bonds.

Cardinality Constraint

$$\sum_{i=1}^{25} y_i \leq K \quad (7)$$

The number of selected assets is limited to K assets.

Linking Constraint

$$x_i \leq M y_i \quad \forall i \in \text{Assets} \quad (8)$$

Investment in an asset is allowed only if it is selected.

Minimum Investment Constraint

$$x_i \geq L y_i \quad \forall i \in \text{Assets} \quad (9)$$

If an asset is selected, a minimum investment must be made.

Nonnegativity and Binary Constraints

$$x_i, x_f \geq 0 \quad \forall i \quad (10)$$

$$y_i \in \{0, 1\} \quad \forall i \quad (11)$$

3 Implementation and Solution Analysis

3.1 Computational Implementation

The portfolio optimization model was implemented in SAS using PROC OPTMODEL, which allows for mixed-integer linear programming (MILP) formulation and solution. The model was built in a structured workflow consisting of data input, benchmark optimization, and iterative goal programming.

Historical price data for all selected assets was gathered from [9]. Then, the transformed asset data (expected returns, betas, and asset classification) were imported from an external Excel dataset and processed within SAS to define the optimization set. Each asset was indexed and matched with its corresponding financial data, allowing the model to be applied across the entire set of assets. The model was then solved using the MILP solver embedded in SAS. This is necessary due to the inclusion of binary decision variables representing asset selection constraints, which transform the problem from a continuous portfolio optimization into a combinatorial optimization problem.

3.2 Benchmark Portfolio Construction

Before implementing the goal programming framework, two benchmark optimization problems were solved:

- A **maximum return portfolio**, which identifies the highest achievable expected return under all investment constraints.
- A **minimum beta portfolio**, which identifies the lowest systematic risk exposure achievable in the feasible region.

These two solutions define the extreme points of the feasible risk–return space, which serve as anchors for constructing the efficient frontier. The difference between these solutions illustrates the trade-off boundary between return maximization and risk minimization under realistic investment constraints.

3.3 Efficient Frontier Generation Procedure

The efficient frontier is generated using a goal programming approach with varying preference weights. Instead of solving a single optimization problem,

the model is repeatedly solved across a grid of weight combinations between return and risk objectives.

In addition to varying the objective weights, multiple target return and target beta pairs are considered. Each pair defines a separate goal programming model, allowing for comparison of how different target levels influence the resulting portfolio solutions.

For each scenario and iteration:

- The target return and target beta values are fixed
- The relative importance of return and risk is adjusted
- The MILP is re-optimized from scratch
- A new optimal portfolio is obtained

This produces multiple sets of Pareto-efficient solutions, each corresponding to a different target configuration and investor preference profile.

The resulting frontiers are not smooth but piecewise in nature due to:

- cardinality constraints limiting the number of assets
- minimum investment thresholds
- binary selection variables restricting continuous adjustment of positions

3.4 Solver Behavior and Computational Observations

The SAS MILP solver employs a branch-and-bound algorithm with presolve reductions, which significantly improves computational efficiency by eliminating redundant constraints and fixing variables before enumeration.

Across iterations, the solver demonstrates:

- Rapid convergence to optimality (often in a single node expansion stage)
- Strong presolve performance reducing problem dimensionality
- Stable objective values for higher return weights, indicating binding constraint regions

In several cases, multiple weight combinations converge to identical portfolio allocations. This occurs because once the model reaches a constraint boundary (particularly diversification and asset limit constraints), further changes in weights do not alter the optimal solution.

Scenario	Target Return (R_0)	Target Beta (β_0)
Baseline	$0.85R_{\max}$	$25\beta_{\min}$
High Return Target	$0.98R_{\max}$	$25\beta_{\min}$
Tighter Beta Target	$0.85R_{\max}$	$5\beta_{\min}$

Table 1: Target return and beta settings used to generate efficient frontiers

3.5 Interpretation of Efficient Frontier Results

Scenario	Return Weight	Beta Weight	Portfolio Beta	Portfolio Return	Risk-Free Investment	Selected Assets	Target Beta
Baseline	0.1	0.9	0.04199	0.03911	80000.00	2	0.32934
Baseline	0.2	0.8	0.32934	0.14245	60527.08	4	0.32934
Baseline	0.3	0.7	0.32934	0.14245	60527.08	4	0.32934
Baseline	0.4	0.6	0.32934	0.14281	60648.11	4	0.32934
Baseline	0.5	0.5	0.32934	0.14281	60648.11	4	0.32934
Baseline	0.6	0.4	0.32934	0.14281	60648.11	4	0.32934
Baseline	0.7	0.3	0.32934	0.14281	60648.11	4	0.32934
Baseline	0.8	0.2	0.34310	0.14756	60000.00	4	0.32934
Baseline	0.9	0.1	0.67961	0.25024	9000.00	11	0.32934
High Return Target	0.1	0.9	0.04199	0.03911	80000.00	2	0.32934
High Return Target	0.2	0.8	0.32934	0.14245	60527.08	4	0.32934
High Return Target	0.3	0.7	0.32934	0.14245	60527.08	4	0.32934
High Return Target	0.4	0.6	0.32934	0.14281	60648.11	4	0.32934
High Return Target	0.5	0.5	0.32934	0.14281	60648.11	4	0.32934
High Return Target	0.6	0.4	0.32934	0.14281	60648.11	4	0.32934
High Return Target	0.7	0.3	0.32934	0.14281	60648.11	4	0.32934
High Return Target	0.8	0.2	0.34310	0.14756	60000.00	4	0.32934
High Return Target	0.9	0.1	0.81889	0.28038	0.00	10	0.32934
Tighter Beta Target	0.1	0.9	0.02635	0.03557	80000.00	3	0.02635
Tighter Beta Target	0.2	0.8	0.02635	0.03557	80000.00	3	0.02635
Tighter Beta Target	0.3	0.7	0.02635	0.03722	79000.00	4	0.02635
Tighter Beta Target	0.4	0.6	0.02635	0.03699	79000.00	4	0.02635
Tighter Beta Target	0.5	0.5	0.02635	0.03699	79000.00	4	0.02635
Tighter Beta Target	0.6	0.4	0.02635	0.03787	77932.98	3	0.02635
Tighter Beta Target	0.7	0.3	0.02805	0.03805	79000.00	3	0.02635
Tighter Beta Target	0.8	0.2	0.34310	0.14756	60000.00	4	0.02635
Tighter Beta Target	0.9	0.1	0.67961	0.25024	9000.00	11	0.02635

Table 2: Efficient frontiers under different target return and beta settings



Figure 1: Efficient frontiers under different target return and beta settings

The generated results across the three different target settings reveal three distinct portfolio regimes. While the specific target return and beta values vary across scenarios, the overall structure of the solutions remains consistent, indicating similar behavior across the models.

3.5.1 Risk-averse regime (low return weight)

When return is assigned low importance, the models heavily favor risk minimization. This results in:

- High allocation to the risk-free asset
- Very limited exposure to risky assets
- Extremely low portfolio beta values

This behavior is consistent across all target pairings, including the tighter beta scenario, where the already conservative allocation becomes even more pronounced. This regime represents conservative investors prioritizing capital preservation.

3.5.2 Transition regime (moderate weights)

At moderate weight values, the portfolios begin to adjust as the importance of return increases:

- Risk-free allocation decreases

- Additional assets are introduced into the portfolio
- Both return and beta increase

Across the baseline and higher return target models, this transition is relatively limited, with portfolios quickly stabilizing after an initial adjustment. In contrast, the tighter beta target model exhibits a more constrained transition, as the stricter risk requirement restricts the set of feasible portfolios. As a result, the trade-off between return and risk is only observed over a narrow range of weights.

3.5.3 Dominant portfolio regime (high return weight)

For higher return preferences, all models converge to stable portfolio solutions:

- Portfolio return and beta stabilize within each scenario
- Asset composition remains unchanged across multiple weight levels

Despite differences in target values, the baseline and higher return target models produce nearly identical portfolios over a wide range of weights, indicating that the same feasible portfolio dominates both scenarios. This suggests that relaxing the return target does not significantly expand the set of optimal solutions.

In the tighter beta target model, the portfolio remains highly conservative for lower weights, but as the return weight increases, the model begins to accept violations of the beta target in order to achieve higher returns. This results in a sharp transition to higher-risk portfolios, after which the solution again stabilizes.

These results indicate that the solutions become insensitive to further changes in the objective weights once a dominant feasible portfolio is identified. This behavior is driven by the interaction between the model constraints and the goal programming formulation. In particular, the cardinality, diversification, and minimum investment constraints significantly restrict the feasible set, while the use of deviation variables allows the models to trade off violations of the beta target when return is heavily prioritized.

3.6 Key Computational Insight

A key observation from the implementation is that the efficient frontier is discrete in nature. Across different target settings, the efficient frontiers exhibit similar piecewise behavior. This is primarily due to integer constraints on asset selection and minimum investment rules, which restrict gradual reallocation between assets.

In addition, the results show that the frontiers are not only discrete, but also characterized by large regions of stability. Across multiple weight combinations

and different target settings, the models frequently converge to identical portfolio allocations. This indicates the presence of dominant feasible portfolios that satisfy the model constraints and approximate the target values effectively.

As a result, changes in preference weights or target parameters do not always produce new solutions. Instead, the optimization exhibits periods of sharp transitions between portfolios, followed by extended intervals in which the solution remains unchanged. This behavior reflects the combined impact of discrete decision variables and restrictive feasibility conditions on the structure of the efficient frontiers.

3.7 Summary

The SAS implementation successfully demonstrates how a mixed-integer goal programming framework can be used to construct a realistic portfolio optimization model under CAPM assumptions. The computational results show that portfolio structure is jointly determined by investor preferences, target parameter choices, and binding real-world constraints.

In particular, the results highlight that the efficient frontiers are shaped not only by the trade-off between return and risk, but also by the interaction between model constraints and goal programming targets. The presence of overlapping frontiers and repeated portfolio solutions across different scenarios indicates that the feasible region is highly restrictive, leading to a limited set of dominant portfolios.

Overall, the model provides insight into how practical investment constraints and target-driven optimization can significantly influence portfolio selection, often limiting the range of achievable trade-offs.

4 Conclusion

We developed a portfolio optimization model based on the Capital Asset Pricing Model (CAPM), formulated as a mixed-integer goal programming problem. By incorporating practical investment constraints, including cardinality limits, diversification requirements, and minimum investment thresholds, the model provides a realistic framework for portfolio selection. The computational results demonstrate that portfolio outcomes are influenced not only by the trade-off between return and systematic risk, but also by the interaction between target parameter choices and binding constraints. In particular, the efficient frontiers generated under different target settings exhibit significant overlap, indicating that the feasible region is highly restrictive and often leads to a limited number of dominant portfolio solutions. Furthermore, the results show that while changes in objective weights can induce transitions between portfolios, large regions of stability exist where the optimal allocation remains unchanged. This behavior reflects the discrete nature of the model and the limited flexibility imposed by real-world investment constraints.

The proposed framework highlights the importance of combining optimization techniques with practical considerations in portfolio construction. While the goal programming approach provides flexibility in modeling investor preferences, the results suggest that constraint structure and target specification play a critical role in determining the range of achievable portfolio outcomes.

Future work may consider extending the model to incorporate alternative risk measures, such as variance or downside risk, as well as dynamic or multi-period portfolio optimization settings.

References

- [1] Markowitz, H. (1952). *Portfolio Selection*. The Journal of Finance, 7(1), 77–91.
- [2] Tobin, J. (1958). *Liquidity Preference as Behavior Towards Risk*. The Review of Economic Studies, 25(2), 65–86.
- [3] Sharpe, W. F. (1964). *Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk*. The Journal of Finance, 19(3), 425–442.
- [4] Lintner, J. (1965). *The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets*. The Review of Economics and Statistics, 47(1), 13–37.
- [5] Mossin, J. (1966). *Equilibrium in a Capital Asset Market*. Econometrica, 34(4), 768–783.
- [6] Black, F., and Litterman, R. (1992). *Global Portfolio Optimization*. Financial Analysts Journal, 48(5), 28–43.
- [7] Charnes, A., Cooper, W. W., and Ferguson, R. O. (1955). *Optimal Estimation of Executive Compensation by Linear Programming*. Management Science, 1(2), 138–151.
- [8] Fama, E. F., and French, K. R. (1993). *Common Risk Factors in the Returns on Stocks and Bonds*. Journal of Financial Economics, 33(1), 3–56.
- [9] Yahoo Finance. *Historical Market Data*. Available at: <https://finance.yahoo.com/>